



澳門理工大學  
Universidade Politécnica de Macau  
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應用科學學院  
Faculdade de Ciências Aplicadas  
Faculty of Applied Sciences

*Doctor of Philosophy in Computer Applied Technology*

**Thesis Defence Examination**

**Thesis Review Comments**

**Student**

|                    |                 |    |          |
|--------------------|-----------------|----|----------|
| Name               | Na Han          | ID | P2314691 |
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| Title of Thesis    |                 |    |          |
| Date of Submission | 2026-06-04      |    |          |

**Reviewer**

|      |              |      |            |
|------|--------------|------|------------|
| Name | Dr. Hong Lin | Date | 2026-06-17 |
|------|--------------|------|------------|

**Recommendation**

☒ Accepted ☐ Suggestion for Revision\*

\* "Suggestion for Revision" will result in the delay of the oral defence.

**Comments**

**Overview.**

This study develops deep learning-based methods for diagnosing lung cancer and thyroid cancer from routine chest CT. For pulmonary nodules, the author proposes CL-IJLO, an end-to-end framework addressing longitudinal data scarcity, feature redundancy, and optimization misalignment. It integrates a contrastive feature compression module with a joint contrastive loss for collaborative optimization. For thyroid cancer, the author proposes a Transformer-based DVT model with transfer learning to handle high CT noise and limited samples. This work systematically validates intelligent multi-cancer screening models, providing a practical pathway toward accurate, cost-effective early diagnosis.

**Knowledge Base and Literature Review.**

The author conducted comprehensive literature view, which includes the overview of medical imaging intelligent analysis technology, application of artificial intelligence in the diagnosis of thyroid cancer with enhanced CT, research progress and application of artificial intelligence in the diagnosis and treatment of pulmonary nodules, and research progress in

lung cancer subtype classification. In addition, the author summarized the gap between the existing works and the application requirements.

### ***Originality and Independence.***

The research originally proposed a Small-sample learning method: A Transformer-based model with secondary transfer learning (natural + medical images) to address thyroid data scarcity and enhance generalization. For utilizing the longitudinal CT images, the author proposes a Sequence purification: A contrastive feature compression module that reduces longitudinal redundancy by 80% while preserving predictive performance and boosting efficiency. In addition, the author proposed a combined contrastive loss (CCL) for end-to-end optimization of feature extraction and classification, prioritizing high recall to meet clinical needs.

### ***Arguments and Reasoning.***

Following a standard deep learning research framework, this thesis addresses the proposed problem by performing data preparation and analysis, designing, training, and validating a deep learning neural network, and verifying the effectiveness of the proposed method through experiments. In the literature review, the author summarizes existing methods and clinical challenges, and proposes corresponding solutions to these issues. In the methods section, the author focuses on the network architecture, loss functions, and training strategies tailored for small-sample scenarios. Finally, the effectiveness of the method is validated through experiments. The paper features clear figures and charts and a logical structure. However, some of the formulas in the paper do not conform to standard conventions.

### ***Evaluation within the Field.***

To evaluate the performance, the author uses the reasonable measure metrics and conducts comprehensive experiments. Based on a high-quality longitudinal dataset of 1165 samples constructed in collaboration with Zhuhai People's Hospital, this CL-IJLO achieves an accuracy for a 3-month period: 0.87, and AUC: 0.94; accuracy for a 6-month period: 0.83, AUC: 0.91. The generalization ability and high sensitivity achieves at accuracy: 0.87, AUC: 0.90, sensitivity: 0.97. On the enhanced CT dataset composed of data from multiple centers in Guangdong and Xinjiang, the DVT achieved an accuracy of 0.96 and an AUROC of 0.97, verifying the feasibility of using conventional chest CT for the preliminary screening of thyroid cancer, and providing a potential low-cost screening solution for resource-limited regions. The author also uses the various tables, charts and figures to illustrate the results clearly.

### ***Completeness and Coherence.***

The structure of the thesis is complete, which meets the requirement of a doctoral thesis. The introduction introduces the background of the research. The Chapter 2 shows a comprehensive literature view. In the methodology part, the author detailedly describes and explains the networks and algorithms. For each methods, the author provides the

experimental setting and results. For each research problems proposed, the author proposed corresponding solutions. The entire thesis keeps coherence.

### ***Conclusion.***

This study focuses on the deep-learning for cancer diagnosis with chest CT images. The author proposes a series of deep learning solutions: a dynamic prediction of the malignant risk of pulmonary nodules, an intelligent screening of thyroid cancer based on conventional chest CT, the exploration of multi-modal fusion for the classification of lung tumor sub-types. This thesis also objectively analyzes the limitations and shortcomings of the current work. In addition, the thesis outlines future research directions, with the aim of providing references for subsequent research. This work shows research and application value, and meets the requirements for a Ph.D. in Computer Applied Technology.

### ***Additional Comments.***